

Serafin Otadoy Gargantiel III

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EDUCATION

B.S., Computer Science | University of Illinois at Chicago (UIC) | May 2026 | GPA: 3.42 | Dean's List (Spring & Fall 2023)
Coursework: Data Structures, Software Design, Algorithms, Data Science, Mobile Development, Software Engineering

TECHNICAL SKILLS

Languages: C++, Python, Java, JavaScript, SQL, HTML/CSS, React **Tools:** VS Code, GitHub, Linux Scripting, Jupyter Notebook
Concepts: Object-Oriented Programming, Concurrent & Multi-Threaded Systems, High-Performance Software, Low-Latency Optimization, Agile/TDD, Database Systems, Probability & Statistics

PROJECTS

Concurrent Thread Race Engine (Gopher Hunting Game) | Java · Multi-Threading · Android
Mar 2026 – Apr 2026

- Designed a high-performance concurrent system with two competing threads — sequential vs. randomized — racing to locate a hidden target across independent grids, achieving sub-100ms race resolution.
- Implemented low-latency thread coordination using Android's Handler/Looper architecture; integrated runtime dangerous-permission gating in compliance with Android's permission model.

Semantic Search Engine over Product Dataset | Python · PostgreSQL · Vector Databases
Sep 2025 – Dec 2025

- Built an embedding-based similarity search system over 75,000+ Amazon products with a team of 3, achieving top-5 recall accuracy and sub-200ms query response at runtime.
- Designed and optimized Python ingestion pipelines and PostgreSQL indexes to support high-throughput natural language lookups with statistical relevance ranking.

Make Or Break — Physics & Disaster Simulator | Unity · JavaScript · 3D Systems
Jan 2026 – May 2026

- Personally implemented earthquake and tornado disaster mechanics in a team of 4, engineering real-time force propagation and structural failure logic across prebuilt 3D environments.

Inventory Management Platform | React.js · UI/UX · GitHub Pages
Feb 2026 – May 2026

- Shipped a production-deployed React web app in a team of 5, applying component architecture and responsive design principles for a structured, user-tested inventory workflow.

EXPERIENCE

Part-Time Lab Researcher | UIC Wearable Technology & Sensory Enhancement Lab
May 2026 – Present

- Engineered a real-time React UI for a BLE-connected intra-oral wearable, optimizing data-to-display latency through direct integration with hardware sensor streams.
- Collaborated with engineers across hardware and software teams to align data flow architecture with research participant workflows.

Backend Developer | UIC SparkHacks Hackathon
Feb 2025

- Built a Python/SQL chatbot backend under 24 hours, designing a relational database with similarity-based query matching to support real-time natural language lookups.

Data Analytics Developer | Global Accelerator Program
Aug 2024 – Nov 2024

- Processed and transformed multiple datasets using Python (pandas) and MySQL, applying statistical methods to surface insights visualized in Tableau across 5+ project assignments.

LEADERSHIP & ACTIVITIES

Summer Bio-Engineering Study Abroad | London & Milan — Wearable Technology in Europe
Jul 2025

- Visited 6+ wearable tech and biomedical companies including NeuBond, ARxVision, and Mapei Sport; attended London Tech Week 2025 engaging founders across AI, hardware, and high-performance sensor systems.

SHPE — Society of Professional Hispanic Engineers & FIA — Filipinos In Alliance | UIC Chapter
Mar 2024 – May 2026

- Represented UIC at SHPE National Convention 2025 (Academic Olympiad), competing and networking with engineers and technology firms across the industry. Athletics Student Board Intern at FIA (Jan–Apr 2025).

Semi-Professional Soccer | Berber City FC (UPSL Premier Division) / Machu Picchu FC (GCSL Major Division)
March 2026 - Current

- Played competitive soccer on two semi-pro rosters; balancing 3 training sessions with weekend games alongside engineering work

CERTIFICATIONS

CodePath Intermediate CyberSecurity (May 2026) | Python Specialist (Dec 2024) | SQL Specialist (Oct 2024) | Intel Sustainability Data Analysis (Nov 2024) | Intercultural Skills (Dec 2024) | GRAMMYs Audience Analysis (Dec 2024) | Coding for Data Track (Dec 2024)